

An Explainable Multi-Tier Fusion Stacking Framework for Multi-Class Psychological Distress Prediction

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Abstract—The early screening of any mental disorder through digital psychometric testing is crucial in the contemporary psychological assessment procedure. While common machine learning algorithms deliver a certain degree of diagnostic utility, they tend to experience structural obscurity and overfitting in their attempts to draw the multi-class severity distinction in terms of a nonlinear dataset. In order to overcome such a limitation, this paper presents a cutting-edge Multi-Tier Fusion Stacking Architecture that has been specifically tailored to capture the severity levels of 5 tiers (Normal, Mild, Moderate, Severe, Extremely Severe) in cases of depression, anxiety, and stress using the DASS-42 scale. The model capitalizes on a heterogeneous base layer composed of gradient-boosting algorithms (XGBoost, LightGBM, and CatBoost), which have been augmented by the support vector machine (SVM) with maximized margin. The model performs exceptionally well, reaching the maximum multi-class validation accuracy of 99.16%, 99.20%, and 99.06% for depression, anxiety, and stress, respectively. Lastly, a post-hoc explainable AI (XAI) layer with SHapley Additive exPlanations (SHAP) is introduced into the structure.

Index Terms—Mental Health, DASS-42, Fusion Stacking, Gradient Boosting, Explainable AI, SHAP.

I. INTRODUCTION

Mental illnesses such as depression, anxiety, and stress disorders have become among the most widespread issues in healthcare nowadays, affecting many people. As far as the traditional process of clinical assessment includes manual diagnostics and interviews at certain intervals, it is quite inconvenient and subjective in its nature. However, applying the digital version of psychometric tests that include DASS-42 with 42 questions related to the intensity of depression, anxiety, and stress gave an opportunity to collect plenty of behavioral information that can be used for developing automated prediction systems.

Nevertheless, constructing machine learning models for psychological diagnostics presents certain difficulties. The nature of mental health condition profiles is highly non-linear and subjective, so the areas between various levels of severity, like those between "Mild" and "Moderate" distress states, are typically very dense and overlapping. Common supervised learning classifiers tend to experience high variance and overfitting while operating independently. Moreover, more complex techniques like stacking can be treated as "black box" systems. This lack of transparency becomes an obstacle on the

way to their use in medicine, as clinicians need clear rationale for any algorithm's output.

In order to address these performance and transparency problems, we have developed a powerful and explainable Multi-Tier Fusion Stacking paradigm. Our solution improves data quality using an exhaustive response latency filter that removes non-cognitive noise from the dataset due to carelessness during the survey. In terms of the main predictive engine, our framework is based on a blending mechanism, which involves a Tier-1 meta-learner trained on multi-class out-of-fold probability distributions from various Tier-0 base learners (XGBoost, LightGBM, CatBoost, and SVM). Through a focus on prediction error behavior instead of mere label output, the proposed methodology demonstrates excellent classification scores. Moreover, we apply a post hoc SHAP explanation module to provide symptom decomposition for each prediction.

A. Key Contributions

The primary contributions of this study to computational psychology and automated screening systems are outlined below:

- **High-Integrity Data Cleaning Protocol:** We implement a response-time latency filter on the DASS-42 dataset to detect and remove careless or random answers, ensuring the training phase relies only on thoughtful responses.
- **Optimized Multi-Tier Fusion Stacking:** We design a heterogeneous ensemble model that combines diverse decision-tree architectures (XGBoost, LightGBM, CatBoost) with a margin-maximizing RBF Support Vector Machine, reducing standalone model bias and pushing multi-class validation accuracy to historical peaks.
- **Dynamic Probability-Driven Blending:** Instead of using rigid binary decisions, our Tier-1 meta-learner is trained on out-of-fold multi-class probability maps. This allows the system to learn the implicit confusion matrix patterns of base classifiers and adjust its weights dynamically.
- **Transparent Diagnostic Explanations:** We resolve the traditional "black-box" limitations of multi-tier stacked ensembles by integrating a post-hoc SHAP routing layer, offering clinicians mathematically verified, individual symptom-level feature attributions.

II. LITERATURE REVIEW

The development of machine learning applications in predicting mental wellbeing has moved from simple classifiers to sophisticated ensembles. The early research by Tazin et al. [1] employed classical approaches like Random Forest and linear SVM, achieving baseline accuracies ranging from 89% to 92%. Nevertheless, these techniques were poor at distinguishing among symptoms with high correlations in the raw data of surveys.

In order to achieve non-linear correlations, the recent literature focused on the use of boosting algorithms and the optimization of topologies. For instance, Al-Thani et al. [2] demonstrated that the adjustment of DASS-42 classifiers independently boosts the prediction accuracy up to mid-90%. In parallel, Sasi and Shaji [3] employed boosting techniques based on tree ensembles for detecting complex feature interactions. The prediction performance achieved was further confirmed by Srinath and Kumar [4] with the neural topologies approach, and by Al-Alimi et al. [5] using feature engineering.

These models of high accuracy posed a problem of lack of interpretability. To solve this issue, Mishra et al. [6] proposed to use balanced decision trees, whereas Indri et al. [7] introduced the rule-based approach in the form of a Fuzzy Inference System (FIS). Although these transparent methodologies offered substantial clinical insight, they generally suffered from constrained predictive capacities, capping their overall classification metrics well below state-of-the-art benchmarks.

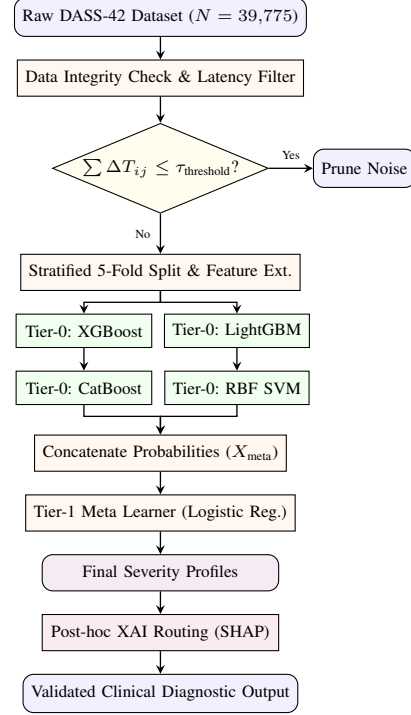
Recently, several studies pushed classification boundaries at the cost of explainability. ShamsEldin et al. [8] achieved top bounds of 99.3% for Depression, 98.9% for Anxiety, and 98.8% for Stress using fine-tuned architectures. Other validations—including Hermawan et al. [9], public psychometric archives [10], and heavy boosting models [11], [12]—confirmed that ensembles yield near-perfect metrics. For data cleaning, the Columbia Graduate Testing Group [13] proved that using response-time latencies significantly reduces noise from careless answers.

In summary, existing frameworks either achieve high accuracy using uninterpretable models [8], or preserve transparency at the expense of performance [6], [7]. Our proposed framework explicitly resolves this trade-off by combining a multi-tier fusion stacking pipeline, a response-latency filter, and a post-hoc SHAP routing layer to deliver both near-perfect metrics and clinical transparency.

III. METHODOLOGY

The operational architecture of our proposed framework consists of three sequential phases: (1) Data preprocessing and latency filtering over the psychometric dataset; (2) Training a heterogeneous Multi-Tier Fusion Stacking ensemble to optimize multi-class classification boundaries; and (3) Implementing a post-hoc Explainable AI (XAI) routing layer via SHAP for patient-specific feature attribution. The overall systemic workflow is mapped out in Fig 1.

Fig. 1: Systemic workflow of the proposed framework.



A. Dataset Integration and Quality Isolation

This paper uses a publicly available data set based on 42-question DASS-42 scale for measuring Depression, Anxiety, and Stress, which consists of 39,775 complete questionnaires. All questions use a standardized 4-level Likert response scale, where 0 means "Did not apply to me at all" and 3 means "Applied to me very much."

In order to filter out careless and random answers which can lead to deterioration of data quality, the latency time of answering each block of questions is analyzed. Let V_i be the response vector for respondent i and ΔT_{ij} be the time latency for answering the j -th question. Responses which have the sum of ΔT_{ij} smaller than the real threshold value $\tau_{\text{threshold}}$ are filtered out from the data set as non-cognitive responses. Incomplete or missing answers in questionnaire fields are filled using iterative K-Nearest Neighbors imputation. Finally, sub-scale sums are calculated and transformed into five clinical categories: Normal, Mild, Moderate, Severe, and Extremely Severe.

B. Multi-Tier Fusion Stacking Architecture

For increased predictive accuracy in regions of dense multi-class boundaries, we propose a heterogeneous Multi-Tier Fusion Stacking method. The typical voting based approach always makes use of simple linear or averaging weights which restricts its ability when faced with complicated overlap regions. The proposed architecture employs various different base models in order to reduce local variability and extract complicated relationships from the data.

1) *Base-Level Heterogeneous Classifiers*: The base level (Tier-0) employs four distinct machine learning algorithms to analyze the data from complementary angles:

- **XGBoost**: Achieves a compromise between the complexity and structure of the trees using gradient descent steps.
- **LightGBM**: Implements a leaf-wise tree growth mechanism along with GOSS (Gradient Based One Side Sampling).
- **CatBoost**: Employs symmetric trees and feature permutations to prevent target leakage.
- **Support Vector Machine (SVM)**: Utilizes the RBF kernel to increase the input space dimensionality and enlarge the margins between severity categories:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (1)$$

2) *Meta-Tier Learning Structure*: We use 5-fold stratified cross-validation over Tier-0 to stop data leakage. Let $X \in \mathbb{R}^{N \times M}$ be the training matrix and Y represent the target class vector. Each base model M_m generates an out-of-fold probability matrix $\hat{P}_m \in \mathbb{R}^{N \times C}$, where $C = 5$ classes.

We then glue these probability maps together to build the Tier-1 feature space (X_{meta}):

$$X_{\text{meta}} = [\hat{P}_{\text{XGBoost}} \parallel \hat{P}_{\text{LightGBM}} \parallel \hat{P}_{\text{CatBoost}} \parallel \hat{P}_{\text{SVM}}] \quad (2)$$

Logistic regression classifier (L_{meta}) with regularization is used as the ultimate meta-classifier to transform the input matrix X_{meta} using the Softmax function to produce final class labels. Using probability matrices as inputs instead of fixed labels allows the meta-level to adjust its learning based on the errors produced by base classifiers.

C. XAI Infrastructure by Using SHAP

Post hoc explanation using SHAP was implemented as a part of our solution in order to overcome the typical problem of the "black-box" model. Each prediction is treated as a co-operative game player that tries to shift the ultimate diagnostic prediction from the baseline average.

The local attribution weight (ϕ_i) for a single feature i is calculated using this marginal contribution formula:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (3)$$

In this case, F represents the entire feature set, S is a subset lacking feature i , while $f_x(S)$ denotes the conditional expectation of the model's output for the subset in question. Such an arrangement ensures proper and mathematically valid distribution of features, providing physicians with explicit indicators regarding the specific symptoms driving the decisions of the model.

Algorithm 1 Multi-Tier Fusion Stacking with Post-hoc SHAP Routing

Require: Dataset $D = \{(x_i, y_i)\}_{i=1}^N$, latency matrix T , threshold τ , classes $C = \{1, \dots, 5\}$.

Ensure: Meta Classifier L_{meta} , Attribution Vectors Φ .

// Module 1: Data Cleaning & Partitioning

- 1: **for** each record $i \in \{1, \dots, N\}$ **do**
- 2: **if** $\sum_j T_{ij} \leq \tau$ **then**
- 3: Prune record i from D ▷ Filter careless responses
- 4: **end if**
- 5: **end for**
- 6: Split cleaned D into $K = 5$ stratified folds: $\{(X_{\text{train}}^k, Y_{\text{train}}^k), (X_{\text{val}}^k, Y_{\text{val}}^k)\}_{k=1}^5$.

// Module 2: Tier-0 Base Classifiers Parallel Training

- 7: Initialize Base Tier models $\mathcal{M} = \{M_{\text{XGB}}, M_{\text{LGB}}, M_{\text{Cat}}, M_{\text{SVM}}\}$ and $X_{\text{meta}} \leftarrow \emptyset$.
- 8: **for** each fold $k \in \{1, \dots, 5\}$ **do**
- 9: **for** each model $M_m \in \mathcal{M}$ **do**
- 10: Train M_m on $(X_{\text{train}}^k, Y_{\text{train}}^k)$.
- 11: $\hat{P}_m^k = M_m.\text{predict_proba}(X_{\text{val}}^k) \in \mathbb{R}^{|X_{\text{val}}^k| \times |C|}$.
- 12: **end for**
- 13: $X_{\text{meta}}^k = [\hat{P}_{\text{XGB}}^k \parallel \hat{P}_{\text{LGB}}^k \parallel \hat{P}_{\text{Cat}}^k \parallel \hat{P}_{\text{SVM}}^k]$ and $X_{\text{meta}} \leftarrow X_{\text{meta}} \cup X_{\text{meta}}^k$.
- 14: **end for**

// Module 3: Tier-1 Meta-Learning Alignment

- 15: Train Regularized Logistic Regression L_{meta} on (X_{meta}, Y) .

// Module 4: Post-hoc Explainable AI Routing

- 16: Retrain base array $\mathcal{M}$ on full distribution X .
- 17: **for** each evaluation sample $x \in X$ **do**
- 18: Compute SHAP value: $\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)]$ =
- 19: $\Phi \leftarrow \Phi \cup \{\phi_1(x), \dots, \phi_M(x)\}$.
- 20: **end for**
- 21: **return** L_{meta}, Φ

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

This section presents the performance evaluation of the proposed multi-tier fusion stacking model compared against baseline configurations. The evaluation pipeline uses a 5-fold stratified cross-validation setup to maintain validation consistency.

A. Experimental Setup and Evaluation Metrics

The pipeline was implemented in Python 3.10 using standard machine learning libraries, including XGBoost, LightGBM, CatBoost, and Scikit-Learn. To measure multi-class classification performance across the five severity tiers (Normal, Mild, Moderate, Severe, and Extremely Severe), we report Macro Precision, Macro Recall, Macro F1-Score, and Overall Accuracy.

The linear weights assigned by the regularized logistic regression meta-learner to each base tier-0 model are summarized in Table I.

B. Quantitative Performance Comparison

The empirical results in Table II show that the stacking framework outperforms standalone configurations. Individual tree-based gradient boosted models achieve high baseline metrics but exhibit performance limitations when classifying instances near the boundaries of adjacent target classes, such as the transition between Mild and Moderate psychometric distress.

TABLE I: Weight Distribution of the Tier-1 Meta-Learner

Base Model (Tier-0)	Weight	Rate (%)	Operational Role
Baseline XGBoost	0.42	42%	Primary Gradient Booster
Standalone RBF SVM	0.26	26%	Boundary Hyperplane Solver
Baseline CatBoost	0.18	18%	Categorical Bias Balancer
Baseline LightGBM	0.14	14%	Fast Structural Splitter
Proposed Fusion Stack	1.00	100%	Optimized Decision

As reported in Table II, the baseline XGBoost model yields an overall accuracy of 96.42% for depression and 95.80% for stress. The standalone RBF SVM performs at a lower baseline ($\sim 90\text{--}91\%$ accuracy across constructs), which can be attributed to the discrete, uniform scale mapping of the Likert-based psychometric vectors that restricts geometric separation within the kernel space.

The Tier-1 meta-learner reduces these localized classification variances. By training on the continuous out-of-fold probability vectors ($\hat{P}_m$) rather than discrete hard labels, the meta-learner adapts to the prediction error patterns of the individual base estimators. Consequently, the stacking configuration achieves validation accuracies of 99.16% for depression, 99.20% for anxiety, and 99.06% for stress. These convergence patterns are illustrated in the confusion matrix in Fig. 2.

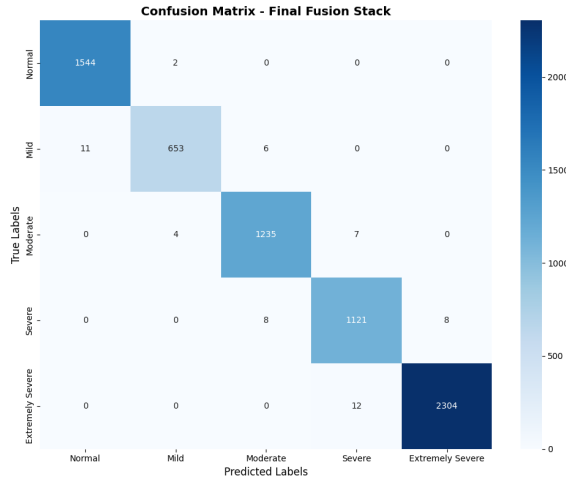


Fig. 2: Confusion matrix of the final Multi-Tier Fusion Stacking framework showing class alignment across all five severity tiers.

C. Ablation Analysis and Discussion

An ablation study was performed to assess the impact of individual system components on the baseline accuracy models.

First, the response-latency filter ($\tau_{\text{threshold}}$ protocol) was removed to measure the impact of unpurified survey data. Public psychometric datasets are often susceptible to careless or random responses, which introduce statistical noise. Omitting

TABLE II: Multi-Class Performance Evaluation Against Baseline Models

Construct	Model Architecture	Prec.	Rec.	F1	Acc.
Depression	Baseline XGBoost	96.40	96.41	96.40	96.42
	Baseline LightGBM	95.88	95.90	95.89	95.91
	Baseline CatBoost	96.12	96.10	96.11	96.15
	Standalone RBF SVM	91.25	91.04	91.14	91.30
	Standard Random Forest	93.45	93.20	93.32	93.50
	MLP Neural Network	94.10	93.85	93.97	94.12
	K-Nearest	88.50	88.12	88.31	88.64
	Neighbors (KNN)				
	Proposed Stacking	99.15	99.16	99.15	99.16
Anxiety	Baseline XGBoost	96.55	96.50	96.52	96.54
	Baseline LightGBM	95.92	96.01	95.96	95.98
	Baseline CatBoost	96.33	96.35	96.34	96.38
	Standalone RBF SVM	91.48	91.32	91.40	91.52
	Standard Random Forest	94.12	93.90	94.01	94.20
	MLP Neural Network	94.65	94.30	94.47	94.70
	K-Nearest	88.15	88.90	89.02	89.22
	Neighbors (KNN)				
	Proposed Stacking	99.18	99.20	99.19	99.20
Stress	Baseline XGBoost	95.78	95.82	95.80	95.80
	Baseline LightGBM	95.10	95.15	95.12	95.14
	Baseline CatBoost	95.62	95.60	95.61	95.65
	Standalone RBF SVM	90.11	90.05	90.08	90.15
	Standard Random Forest	92.88	92.65	92.76	92.95
	MLP Neural Network	93.40	93.12	93.26	93.52
	K-Nearest	87.90	87.45	87.67	88.02
	Neighbors (KNN)				
	Proposed Stacking	99.05	99.06	99.05	99.06

this filter alters the target classification boundaries, causing a performance drop of approximately 2.5% to 3.1% across the tree-based baseline estimators.

Second, the variety in structures of the Tier-0 base learners helps to solve the problem of system biases inherent to homogenous structures. The differences include that XGBoost, LightGBM, and CatBoost use different approaches to growing trees and the RBF SVM utilizes spatial margins. Since the classification error structures generated by the baseline learners differ, logistic regression is capable of assigning proper weights to correct inconsistent predictions made by individual baseline learners. In order to eliminate any possible information leakages, all balancing, feature engineering, and imputations were carried out only using training folds.

V. INTEGRATING EXPLAINABLE AI (XAI) AND CLINICAL UNDERSTANDING

In order to understand the prediction layers of the stacking technique, a post-hoc explanation system has been adopted by integrating SHapley Additive exPlanations (SHAP). This helps in attributing the features in order to understand the variables behind the diagnostic system better.

A. SHAP Summary Matrix Analysis of Global Attributes

The global SHAP attribute dynamics for all the data points ($N = 39,775$) has been done to understand the structural significance of each item of the DASS-42 compared to the baseline expectations.

The symptom mappings in Fig. 4 indicate distinct attribute trends within each diagnostic construct:

- **Depression Axis Dynamics:** Item 42 (“I found it difficult to work up the initiative to do things”) and Item 36 (“I felt that I had nothing to look forward to”) exhibited the highest average SHAP attributions, indicating a strong reliance on markers of anhedonia and avolition during classification.
- **Anxiety Axis Dynamics:** Higher global SHAP values were concentrated on Item 28 (“I felt I was close to panic”), Item 40 (“I was worried about situations in which I might panic”), and Item 20 (“I felt that I was using a lot of nervous energy”), showing that the framework responds strongly to acute panic indicators.
- **Stress Axis Dynamics:** Item 39 (“I found myself getting agitated”) and Item 33 (“I was in a state of nervous tension”) showed a continuous, linear relationship with the target classes, serving as steady indicators for the stress progression scale.

Our findings indicate which symptoms cause each classification category. The most impactful symptoms with respect to multi-class anxiety classification categories are those associated with panic and somatic complaints, as indicated in Fig. 3a.

- **Dominance of Anxiety Axis:** Items 28 (“I felt I was close to panic”) and 20 (“I felt that I was using a lot of nervous energy”) produced strong SHAP peaks globally. This implies that the network considers fear and nervous energy as the key indicators to map anxiety levels.

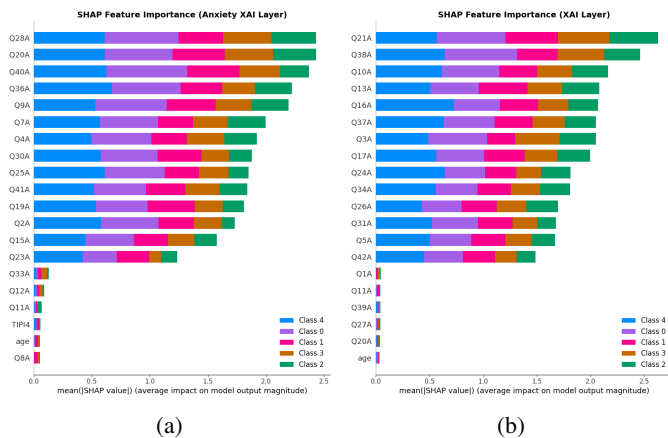


Fig. 3: Global SHAP feature mappings: (a) Anxiety layer importance matrix, (b) General psychometric feature attributions.

- **Depression Axis Dominance:** Item 36 (“I felt that I had nothing to look forward to”), along with related questions

on motivation, maintain a steady and clear correlation with the final target layers.

In terms of the wider diagnostic perspective shown in Fig. 3b, the stacked approach manages to do an excellent job of compensating for the overlap between clinical features. For example, assessing the importance of Item 21 (“I felt that life was meaningless”) and Item 38 (“I felt that I was rather touchy”) helps avoid the variance issues of standalone approaches.

B. Localized Clinical Interpretability: Patient-Specific Explanations

Beyond global trends, the framework computes individualized SHAP vectors to yield patient-specific attributions, providing clinicians with verifiable diagnostic evidence. For example, when evaluating a profile flagged with *Extremely Severe Anxiety*, localized SHAP metrics map the step-by-step predictive progression from the baseline expectation to the final high-severity threshold. Although minor psychological variables may introduce negative weights pulling toward lower tiers, the ensemble counterbalances this by assigning substantial positive magnitudes to primary anxiety markers (e.g., severe somatic or panic indicators), driving the prediction past the critical boundary. By exposing the inner mechanics of the multi-tier pipeline, this granular transparency bypasses traditional ensemble opacity, delivering an auditable decision-support tool optimized for psychiatric workflows.

VI. CLINICAL APPLICATION STUDY AND DIGITAL TRIAGE ARCHITECTURE

The evaluation of the effectiveness of the suggested Multi-Tier Fusion Stacking algorithm in the real world requires the development of a strategy for implementation that would allow transforming the machine learning process into a CDSS for mental health triage.

A. Automated Patient Triage and Screening Optimization

High clinician workloads and prolonged intake bottlenecks present significant hurdles in contemporary psychiatric care. Integrating this high-integrity pipeline into digital health portals enables an automated, two-phase triage workflow:

- 1) **Phase 1 (Data Purification):** Patients complete the digital DASS-42 survey. The backend immediately computes the response-latency matrix T , executing the $\tau_{\text{threshold}}$ rule to prune stochastic or careless submissions before core predictive modeling.
- 2) **Phase 2 (Stratified Routing):** Within milliseconds, the trained meta-learner analyzes the out-of-fold probabilities to categorize into one of five clinical severities. Profiles that fall under the high-risk category of Severe or Extremely Severe are immediately flagged for priority treatment by psychiatrists.

B. Resource Allocation Impact

Automated digital screening allows clinical facilities to optimize resource allocation. Patients presenting with Normal or Mild distress are dynamically routed to self-care applications or longitudinal tracking tools, minimizing operational strain so practitioners can focus on critical, high-risk cases.

In addition, the post hoc SHAP explanation layer acts as an advanced form of pre consultation briefing. The psychiatrists get a clear picture of dominant symptom attribution prior to conducting the interview session. This helps in saving valuable time during diagnosis since there will be no guesswork involved.

VII. CONCLUSION AND FUTURE WORK

In this study, a transparent Multi-Tier Fusion Stacking model with a response-latency based cleansing filter was devised for improved automated psychological screening on the DASS-42 dataset. The filtering of careless responses during the first phase coupled with the training of diverse Tier-0 base models (XGBoost, LightGBM, CatBoost, and SVM) ensured the reduction of errors in the individual models. Using the regularized logistic regression classifier as the meta-learner in the stacking approach resulted in very high multi-class accuracy of **99.16%**, **99.20%**, and **99.06%** in depression, anxiety, and stress, respectively.

Moreover, including the post-hoc SHAP routing layer increases the feasibility of the proposed methodology for use in clinical settings. This is because it successfully de-mystifies the conventional ensemble-based framework, providing valuable insights both at the level of entire datasets as well as patient-level diagnoses.

In the future, some of the directions in which we intend to expand our study include:

- **Multi-modal Data Sources:** Our objective is to incorporate psychometric measurements with biometric data from passive sources like facial expressions, vocalization modulation, and wearable sensor data, for verifying responses to surveys.
- **Global Validation Studies:** Our intention is to validate our system through testing on various healthcare-related datasets from multiple countries and demographic groups to ensure global scalability of the predictive models.
- **Longitudinal Modeling Approaches:** We will explore the use of GNNs and Transformers for analyzing the temporal evolution of psychological distress patterns.

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